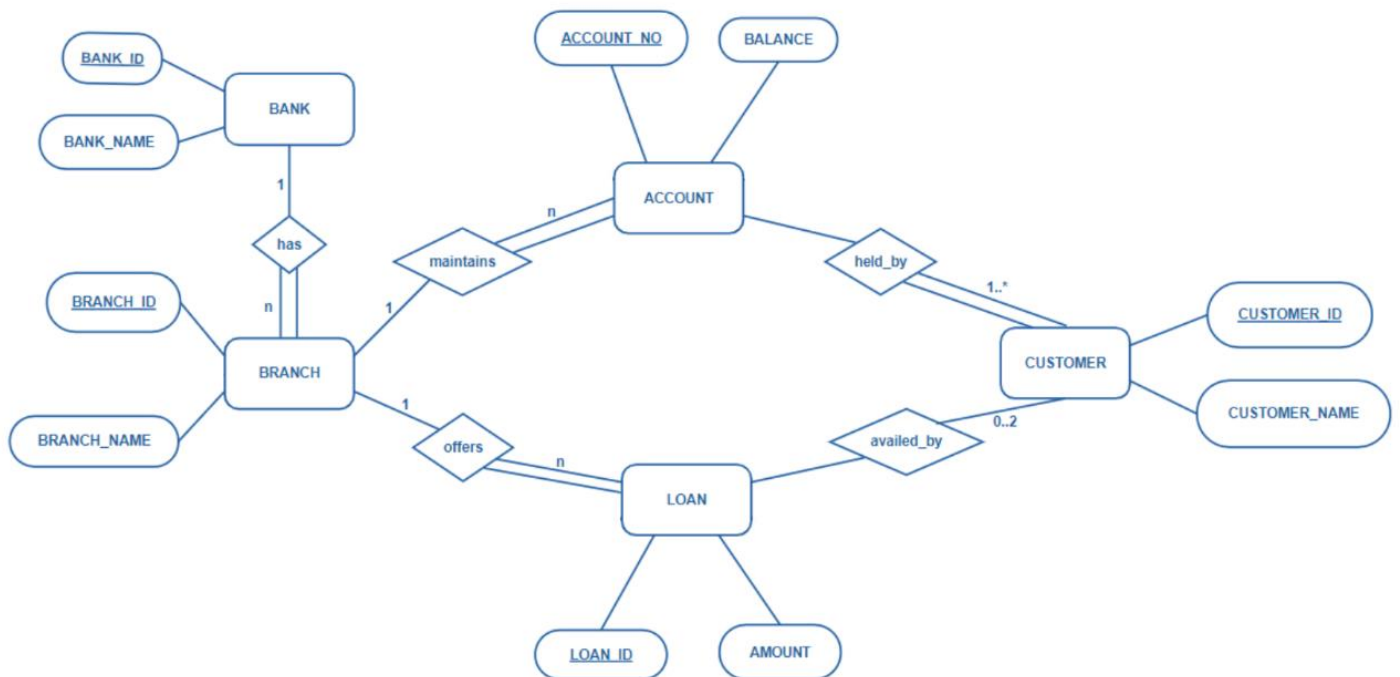


SOUGATA SAHA
IT-A2-037 DBMS ASSIGNMENT 4

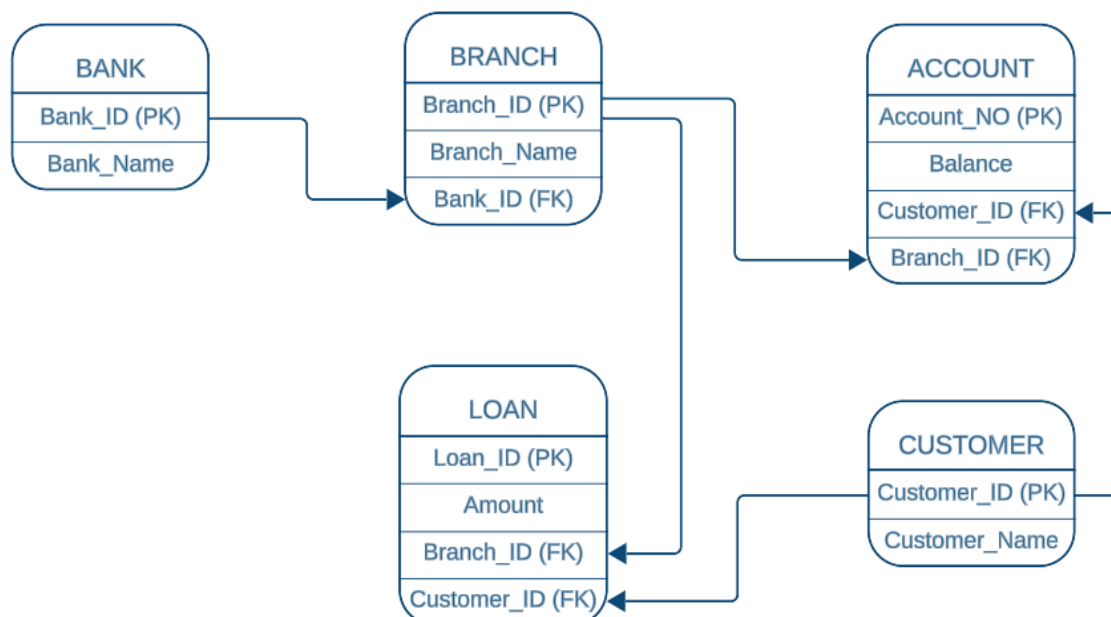
Question:

Consider a BANK database. Each bank can have multiple branches, and each branch can have multiple accounts and loans. Assumptions also can be made. Design an ER diagram and database schema for the system. Specify the primary key, foreign key and other constraints for all required tables. Draw the ER diagram in MS Word.

Er Diagram



Database Schema:



1. Insert at least five tuples in each table.

-- BANK Table

```
Create Table BANK (  
BANK_ID number (2) PRIMARY KEY,  
BANK_NAME varchar (100) NOT NULL);
```

Insert into BANK values (10,'HDFC');

Insert into BANK values (20, 'SBI');

Insert into BANK values (30, 'AXIS');

Insert into BANK values (40, 'PNB');

Insert into BANK values (50, 'ICICI');

-- BRANCH Table

```
Create Table Branch (  
BRANCH_ID number (3) PRIMARY KEY,  
BRANCH_NAME varchar (100) NOT NULL,  
BANK_ID number (2),  
Foreign key (BANK_ID) REFERENCES bank (BANK_ID) on DELETE CASCADE);
```

Insert into BRANCH values (101, 'RUBY', 10);

Insert into BRANCH values (102, 'SEALDAH', 10);

Insert into BRANCH values (103, 'BEHALA', 10);

Insert into BRANCH values (201, 'GARIA', 20);

Insert into BRANCH values (202, 'SEALDAH', 20);

Insert into BRANCH values (203, 'DUMDUM', 20);

Insert into BRANCH values (301, 'BELGACHIA', 30);

Insert into BRANCH values (302, 'PHOOLBAGAN', 30);

Insert into BRANCH values (303, 'RUBY', 30);

Insert into BRANCH values (401, 'BOWBAZAR', 40);

Insert into BRANCH values (402, 'BEHALA', 40);

Insert into BRANCH values (403, 'SEALDAH', 40);

Insert into BRANCH values (501, 'LAKETOWN', 50);

Insert into BRANCH values (502, 'BEHALA', 50);

Insert into BRANCH values (503, 'NEWTOWN', 50);

-- CUSTOMER Table

```
Create Table CUSTOMER (  
CUSTOMER_ID number (5) PRIMARY KEY,  
CUSTOMER_NAME varchar (100) NOT NULL);
```

Insert into CUSTOMER values (10101, 'ABHAY');

Insert into CUSTOMER values (10201, 'SHAHANA');

Insert into CUSTOMER values (10301, 'MANISH');

Insert into CUSTOMER values (20201, 'STELLA');

```
Insert into CUSTOMER values (20202, 'AHANA');
Insert into CUSTOMER values (20101, 'SCARLET');
Insert into CUSTOMER values (20301, 'JOHN');
Insert into CUSTOMER values (30201, 'NEERAJ');
Insert into CUSTOMER values (30101, 'BELLA');
Insert into CUSTOMER values (40101, 'JATIN');
Insert into CUSTOMER values (40201, 'PIYUSH');
Insert into CUSTOMER values (40301, 'SANGITA');
Insert into CUSTOMER values (50101, 'ANWESHA');
Insert into CUSTOMER values (50301, 'PIYALI');
```

-- ACCOUNT Table

```
Create Table ACCOUNT (
ACCOUNT_NO number (11) PRIMARY KEY,
BALANCE decimal (10, 2),
CUSTOMER_ID number (5),
BRANCH_ID number (3),
Foreign key (CUSTOMER_ID) REFERENCES Customer (CUSTOMER_ID) on DELETE CASCADE,
Foreign key (BRANCH_ID) REFERENCES Branch (BRANCH_ID) on DELETE CASCADE);
```

```
Insert into ACCOUNT values (55690129329, 35486.12, 10101, 101);
Insert into ACCOUNT values (89545237546, 1234.56, 10201, 101);
Insert into ACCOUNT values (54321987654, 98765.43, 10201, 102);
Insert into ACCOUNT values (39800005552, 45678.90, 10301, 103);
Insert into ACCOUNT values (59498307311, 2345.57, 20201, 202);
Insert into ACCOUNT values (65592459895, 67890.12, 20202, 202);
Insert into ACCOUNT values (71875151611, 54321.98, 20101, 201);
Insert into ACCOUNT values (77975635172, 8765.43, 20301, 203);
Insert into ACCOUNT values (84531363466, 34567.89, 30101, 302);
Insert into ACCOUNT values (46372949259, 78901.23, 30201, 301);
Insert into ACCOUNT values (82273579151, 65432.10, 40101, 401);
Insert into ACCOUNT values (57679662463, 4321.98, 40201, 402);
Insert into ACCOUNT values (90680460457, 21098.76, 40301, 403);
Insert into ACCOUNT values (18567978674, 3456.78, 40301, 403);
Insert into ACCOUNT values (94720840720, 67789.01, 50101, 501);
Insert into ACCOUNT values (10346725987, 54321.10, 50301, 503);
Insert into ACCOUNT values (35487204793, 9876.54, 50301, 503);
Insert into ACCOUNT values (28645816408, 123456.67, 50301, 503);
```

-- LOAN Table

```
Create Table LOAN (
LOAN_ID number (6) PRIMARY KEY,
LOAN_AMOUNT decimal (12, 2),
CUSTOMER_ID number (5),
```

BRANCH_ID number (3),

Foreign key (CUSTOMER_ID) REFERENCES Customer (CUSTOMER_ID) on DELETE CASCADE,

Foreign key (BRANCH_ID) REFERENCES Branch (BRANCH_ID) on DELETE CASCADE);

Insert into LOAN values (101011, 35486.12, 10101, 101);

Insert into LOAN values (102011, 1234.56, 10201, 101);

Insert into LOAN values (102012, 98765.43, 10201, 102);

Insert into LOAN values (202011, 2345.57, 20201, 202);

Insert into LOAN values (202021, 67890.12, 20202, 202);

Insert into LOAN values (201011, 54321.98, 20101, 201);

Insert into LOAN values (203011, 8765.43, 20301, 203);

Insert into LOAN values (302011, 34567.89, 30201, 302);

Insert into LOAN values (301011, 78901.23, 30101, 301);

Insert into LOAN values (401011, 65432.10, 40101, 401);

Insert into LOAN values (403011, 21098.76, 40301, 403);

Insert into LOAN values (403012, 3456.78, 40301, 403);

Insert into LOAN values (501011, 67789.01, 50101, 501);

Insert into LOAN values (503011, 54321.10, 50301, 503);

Insert into LOAN values (503012, 9876.54, 50301, 503);

Select * from BANK

BANK_ID	BANK_NAME
10	HDFC
20	SBI
30	AXIS
40	PNB
50	ICICI

Select * from BRANCH

BRANCH_ID	BRANCH_NAME	BANK_ID
101	RUBY	10
102	SEALDAH	10
103	BEHALA	10
201	GARIA	20
202	SEALDAH	20
203	DUMDUM	20
301	BELGACHIA	30
302	PHOOLBAGAN	30
303	RUBY	30
401	BOMBAZAR	40
402	BEHALA	40
403	SEALDAH	40
501	LAKETOWN	50
502	BEHALA	50
503	NEWTOWN	50

Select * from CUSTOMER

CUSTOMER_ID	CUSTOMER_NAME
10101	ABHAY
10201	SHAHANA
10301	MANISH
20201	STELLA
20202	AHANA
20101	SCARLET
20301	JOHN
30201	NEERAJ
30101	BELLA
40101	JATIN

Select * from ACCOUNT

ACCOUNT_NO	BALANCE	CUSTOMER_ID	BRANCH_ID
55690129329	35486.12	10101	101
89545237546	1234.56	10201	101
54321987654	98765.43	10201	102
39800005552	45678.9	10301	103
59498307311	2345.57	20201	202
65592459895	67890.12	20202	202
71875151611	54321.98	20101	201
77975635172	8765.43	20301	203
84531363466	34567.89	30101	302
46372949259	78901.23	30201	301
82273579151	65432.1	40101	401
57679662463	4321.98	40201	402
90680460457	21098.76	40301	403
18567978674	3456.78	40301	403
94720840720	67789.01	50101	501
10346725987	54321.1	50301	503
35487204793	9876.54	50301	503
28645816408	123456.67	50301	503

Select * from LOAN

LOAN_ID	LOAN_AMOUNT	CUSTOMER_ID	BRANCH_ID
101011	35486.12	10101	101
102011	1234.56	10201	101
102012	98765.43	10201	102
202011	2345.57	20201	202
202021	67890.12	20202	202
201011	54321.98	20101	201
203011	8765.43	20301	203
302011	34567.89	30201	302
301011	78901.23	30101	301
401011	65432.1	40101	401
403011	21098.76	40301	403
403012	3456.78	40301	403
501011	67789.01	50101	501
503011	54321.1	50301	503
503012	9876.54	50301	503

2. Every customer must have at least one account but is restricted to at most two loans at a time.

Alter table CUSTOMER

Add ACCOUNT_NO number (11) NOT NULL

Add constraint FK foreign key (ACCOUNT_NO) REFERENCES account (ACCOUNT_NO);

Create TRIGGER MaxLoansPerCustomer

Before insert on LOAN

For each row

BEGIN

DECLARE loan_count INT;

Select COUNT (*) INTO loan_count

from LOAN

where CUSTOMER_ID = NEW.CUSTOMER_ID;

IF loan_count >= 2 THEN

SIGNAL SQLSTATE '45000'

SET MESSAGE_TEXT = 'A customer can have a maximum of 2 loans.';

END IF;

END

Triggers	
Trigger	Body
MAXLOANSPERCUSTOMER ENABLED INSERT	<pre> BEGIN DECLARE loan_count INT; Select COUNT (*) INTO loan_count from LOAN where CUSTOMER_ID = NEW.CUSTOMER_ID; IF loan_count >= 2 THEN SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'A customer can have a maximum of 2 loans.'; END IF; END; </pre>

3. Give all the account details of a person who has accounts in SBI.

```

Select A.*
from ACCOUNT A
join BRANCH B ON A.Branch_ID = B.Branch_ID
join BANK K ON B.Bank_ID = K.Bank_ID

```

```

Select A.*
from ACCOUNT A
join BRANCH B ON A.Branch_ID = B.Branch_ID
join BANK K ON B.Bank_ID = K.Bank_ID
where K.Bank_Name = 'SBI'

```

ACCOUNT_NO	BALANCE	CUSTOMER_ID	BRANCH_ID
71875151611	54321.98	20101	201
59498307311	2345.57	20201	202
65592459895	67890.12	20202	202
77975635172	8765.43	20301	203

where K.Bank_Name = 'SBI';

4. Find the account holder name who has more than 2 accounts.

```

Select CUSTOMER_NAME
from CUSTOMER
where CUSTOMER_ID in (
    select A. CUSTOMER_ID
    from ACCOUNT A
    group by A. CUSTOMER_ID
    having COUNT(A.ACCOUNT_No) > 2);

```

```

Select CUSTOMER_NAME
from CUSTOMER
where CUSTOMER_ID in (
  select A. CUSTOMER_ID
  from ACCOUNT A
  group by A. CUSTOMER_ID
  having COUNT(A.ACCOUNT_No) > 2
)

```

CUSTOMER_NAME
PIYALI

5. Rename the accounts table as account details.

Alter table ACCOUNT Rename to ACCOUNT_DETAILS;

```

Alter table ACCOUNT Rename to ACCOUNT_DETAILS

```

Table altered.

6. Find the loan amount and loan taken from which bank for each account holder.

Select C.Customer_Name, L.Loan_Amount, B.Bank_Name
from LOAN L

join CUSTOMER C on L.Customer_ID = C.Customer_ID

join BRANCH BR on L.Branch_ID = BR.Branch_ID

join BANK B on BR.Bank_ID = B.Bank_ID;

```

Select C.Customer_Name, L.Loan_Amount, B.Bank_Name
from LOAN L
join CUSTOMER C on L.Customer_ID = C.Customer_ID
join BRANCH BR on L.Branch_ID = BR.Branch_ID
join BANK B on BR.Bank_ID = B.Bank_ID

```

CUSTOMER_NAME	LOAN_AMOUNT	BANK_NAME
ABHAY	35486.12	HDFC
SHAHANA	1234.56	HDFC
SHAHANA	98765.43	HDFC
STELLA	2345.57	SBI
AHANA	67890.12	SBI
SCARLET	54321.98	SBI
JOHN	8765.43	SBI
NEERAJ	34567.89	AXIS
BELLA	78901.23	AXIS

7. Find the account no. and account holder name who has not taken any loan.

```
Select A.Account_No, C.Customer_Name
from ACCOUNT_DETAILS A
join CUSTOMER C on A.Customer_ID = C.Customer_ID
left join LOAN L on A.Customer_ID = L.Customer_ID
where L.Loan_ID IS NULL
```

```
Select A.Account_No, C.Customer_Name
from ACCOUNT_DETAILS A
join CUSTOMER C on A.Customer_ID = C.Customer_ID
left join LOAN L on A.Customer_ID = L.Customer_ID
where L.Loan_ID IS NULL
```

ACCOUNT_NO	CUSTOMER_NAME
3980005552	MANISH
57679662463	PIYUSH

8. Delete the account of all the persons who had accounts in PNB, Sealdah branch.

```
Delete from ACCOUNT_DETAILS
where BRANCH_ID =
  (Select Branch_ID from BRANCH where BRANCH_NAME = 'Sealdah'
   and BANK_ID = (Select BANK_ID from BANK where BANK_NAME = 'PNB'));
```

```
Delete from ACCOUNT_DETAILS
where BRANCH_ID =
  (Select Branch_ID from BRANCH where BRANCH_NAME = 'Sealdah'
   and BANK_ID = (Select BANK_ID from BANK where BANK_NAME = 'PNB'))

0 row(s) deleted.
```

9. Update the branch to SBI, Salt Lake branch for all the persons who had a SBI account in Sealdah branch.

```
Update BRANCH
set BRANCH_NAME = 'SALT LAKE'
where BANK_ID = (select BANK_ID from BANK where BANK_NAME = 'SBI')
and BRANCH_NAME = 'SEALDAH';
```

10. Find the maximum account balance of a person with account no 54321987654 among all of

```
Update BRANCH
set BRANCH_NAME = 'SALT LAKE'
where BANK_ID = (select BANK_ID from BANK where BANK_NAME = 'SBI')
and BRANCH_NAME = 'SEALDAH'

1 row(s) updated.
```

10. Find the maximum account balance of a person with account no 54321987654 among all of his accounts.

Select MAX(Balance)

from ACCOUNT_DETAILS

where CUSTOMER_ID =
(select CUSTOMER_ID from ACCOUNT_DETAILS where ACCOUNT_NO = 54321987654);

```
Select MAX(Balance)
from ACCOUNT_DETAILS
where CUSTOMER_ID =
(select CUSTOMER_ID from ACCOUNT_DETAILS where ACCOUNT_NO = 54321987654)
```

MAX(BALANCE)

98765.43
